

Qn	Suggested Solutions
6(a)	The magnetic flux density of a magnetic field is the force per unit length acting on a straight, current-carrying conductor,
	carrying unit current and placed at right angles to this external magnetic field.
(b)(i)	The wire exerts a downward force on the magnet.
	By Newton's third law, the magnet exerts an upward force on the wire.
(ii)	A to B
(iii)	$F = BIl \sin \theta$ $= BIl \sin(90^\circ)$ $= (3.7 \times 10^{-3})(4.6)(8.5 \times 10^{-2})$
	$F = 1.447 \times 10^{-3}$ $\approx 1.4 \times 10^{-3} \text{ N}$
(iv)	Assume θ is the angle of rotation from the initial position. length of wire in magnetic field = $\frac{l}{\cos \theta}$
	$F = BI \left(\frac{l}{\cos \theta} \right) [\sin(90^\circ - \theta)]$ $F = BI \left(\frac{l}{\cos \theta} \right) (\cos \theta)$ $F = BIl$ The force acting on the wire is independent of the angle of rotation.

Anglo-Chinese Junior College

H2 (9749) Physics

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